

Documentation Report

On

AnemiaFusionNet: A Multimodal Feature Fusion Framework for Region-Aware Anemia Detection

1. Project Title

AnemiaFusionNet: A Multimodal Feature Fusion Framework for Region-Aware Anemia Detection

2. Abstract

Anemia is a common health condition caused by a reduction in hemoglobin levels, affecting millions of people worldwide. Traditional anemia diagnosis mainly depends on blood tests, which may not always be easily accessible in resource-limited regions.

This project proposes **AnemiaFusionNet**, a multimodal Artificial Intelligence-based anemia detection system that combines three different information sources: **conjunctiva eye images, clinical information, and geographical health risk factors**.

The system uses a Convolutional Neural Network (CNN) to extract visual features from conjunctiva images. Clinical and demographic information is processed using a machine learning-based feature extraction model. Additionally, a Geo-Risk Module incorporates district-level health indicators from NFHS-5 data to capture regional anemia risk patterns.

The extracted features from all three modalities are combined using a **Transformer-based Multimodal Fusion Network** to improve prediction performance. A Streamlit-based application was developed to provide an interactive interface for anemia risk prediction.

The proposed framework demonstrates the effectiveness of multimodal learning by integrating medical images, patient information, and geographical factors into a single AI-based anemia detection system.

3. Introduction

Anemia detection is an important healthcare challenge, especially in areas where laboratory-based diagnosis is difficult. Artificial Intelligence provides an opportunity to develop faster and more accessible screening solutions.

Single-modality approaches such as image-only or clinical-only models may ignore important information available from other sources. Therefore, this project introduces a multimodal approach where multiple data sources are combined to improve prediction reliability.

The proposed system, AnemiaFusionNet, uses:

- Conjunctiva image analysis
- Clinical feature analysis
- Region-aware Geo-Risk analysis

and combines them using Transformer-based feature fusion.

4. Objectives

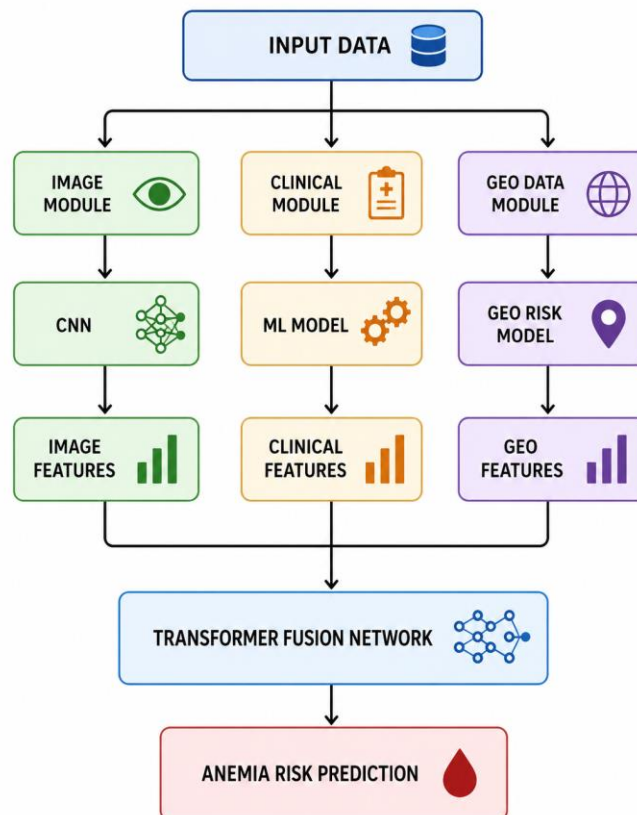
The main objectives of this project are:

1. To develop an AI-based anemia detection system using multiple data modalities.
2. To build a CNN-based image analysis model for conjunctiva images.
3. To process clinical and demographic information using machine learning techniques.
4. To incorporate geographical anemia risk information using NFHS-5 district-level data.
5. To design a Transformer-based multimodal fusion architecture.
6. To develop a Streamlit application for real-time anemia risk prediction.

5. System Architecture

The proposed system consists of three independent feature extraction modules followed by a fusion network.

Overall Workflow



6. Dataset Description

6.1 Image Dataset

The image dataset contains conjunctiva eye images collected from:

- India Dataset
- Italy Dataset

The dataset contains:

- Original conjunctiva images
- Forniceal conjunctiva images
- Palpebral conjunctiva images
- Combined conjunctiva images

The images are used for training the image feature extraction model.

6.2 Clinical Dataset

Clinical information contains patient-level attributes including:

- Hemoglobin level (Hgb)
- Gender
- Age
- Country

The clinical dataset is used to identify anemia-related patterns.

6.3 Geographical Dataset

The Geo-Risk module uses district-level health indicators from NFHS-5 data.

The dataset contains:

- State information
- District information
- Demographic indicators
- Health indicators
- Lifestyle factors
- Nutrition-related features

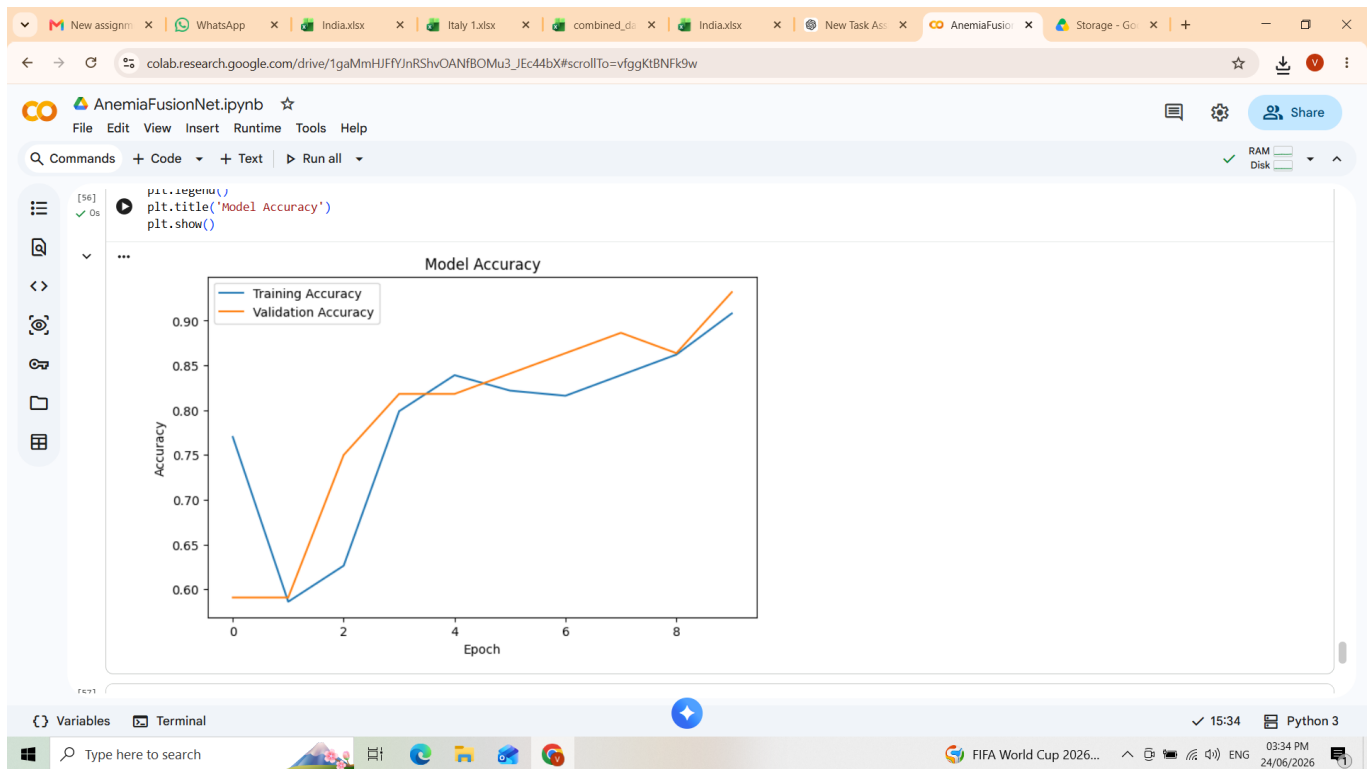
Total geo features used:

109 features

The module provides region-aware anemia risk information.

Performance Evaluation

The training and validation curves indicate that the model successfully learned relevant features from conjunctiva images. The loss decreased over epochs while accuracy improved, demonstrating effective convergence of the CNN model.



7. Methodology

7.1 Image Modality

A Convolutional Neural Network (CNN) model is used for extracting visual information from conjunctiva images.

Steps:

1. Image loading
2. Image resizing
3. Normalization
4. CNN feature extraction
5. Classification output

The model learns visual patterns associated with anemia conditions.

7.2 Clinical Modality

Clinical data is processed using a machine learning model.

Input features:

- Hemoglobin
- Gender
- Age
- Country

The model learns relationships between patient information and anemia status.

7.3 Geo-Risk Module

The Geo-Risk Module introduces location awareness into the system.

The user selects:

- State
- District

The system retrieves corresponding NFHS-5 district information and predicts regional anemia risk.

This improves the model by considering environmental and population-level factors.

7.4 Multimodal Fusion

The outputs of all three modules are combined.

The fusion network uses a Transformer architecture.

The Transformer:

- Learns relationships between different modalities
- Assigns importance to different features
- Produces a combined representation

The final output predicts anemia risk.

8. Model Implementation

Technologies Used

Programming Language:

- Python

Libraries:

- TensorFlow / Keras
- Scikit-learn
- Pandas
- NumPy
- Streamlit

9. Developed Models

Image Model

Architecture:

- CNN-based deep learning model

Purpose:

Extract visual features from conjunctiva images.

Saved Model:

`image_model.keras`

Clinical Model

Algorithm:

Random Forest Classifier

Input:

Clinical and demographic information

Saved Model:

`clinical_model.pkl`

Geo-Risk Model

Algorithm:

Random Forest Regressor

Input:

NFHS-5 district-level indicators

Saved Model:

`geo_model.pkl`

Transformer Fusion Model

Architecture:

Transformer Encoder-based multimodal fusion network

Input:

- Image features
- Clinical features
- Geo features

Saved Model:

`final_anemia_fusion_model.keras`

10. Results and Performance

Image Model

The CNN model successfully learned visual patterns from conjunctiva images.

Clinical Model

Performance:

Accuracy:

100%

The clinical model successfully classified anemia based on patient information.

Geo-Risk Model

Performance:

MAE:

3.38

R² Score:

0.966

The model effectively captured regional health risk patterns.

Fusion Model

Transformer Fusion Training Results:

Training Accuracy:

Approximately 89%

Validation Accuracy:

Approximately 75%

The multimodal fusion model successfully combined information from all three sources.

11. Streamlit Application

A user-friendly web application was developed.

The application provides:

Input:

1. Upload conjunctiva image
2. Enter clinical details
3. Select:
 - State
 - District

Output:

- Anemia risk prediction
- Fusion score

AnemiaFusionNet

localhost:8501

Deploy

AnemiaFusionNet

A Multimodal Feature Fusion Framework for Region-Aware Anemia Detection



About this Project

AnemiaFusionNet is an AI-powered anemia risk screening system that combines **three independent data sources** to produce a more reliable prediction than any single source alone:

- Conjunctiva Eye Image — analyzed using a deep learning (CNN/ViT) model

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Feels hotter

06:37 PM
25/06/2026

AnemiaFusionNet

localhost:8501

Deploy



About this Project

AnemiaFusionNet is an AI-powered anemia risk screening system that combines **three independent data sources** to produce a more reliable prediction than any single source alone:

- Conjunctiva Eye Image — analyzed using a deep learning (CNN/ViT) model
- Clinical & Demographic Data — hemoglobin level, age, gender, and location
- Geo-Risk Module — state and district-level anemia prevalence derived from NFHS-5 government survey data

These three modalities are processed independently and then merged using a **Transformer-based Fusion Network**, which produces a single, region-aware anemia risk score.

> Why Multimodal Fusion?

This tool is for educational/screening purposes only and is not a substitute for professional medical diagnosis.

Start Anemia Risk Assessment

Type here to search

Feels hotter

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AnemiaFusionNet

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Step 1 of 1 — Provide your information across all three modalities

← Back to Welcome

1

Image Modality — Conjunctiva Eye Image

Upload a clear photo of the inner eyelid (conjunctiva)

Drag and drop file here

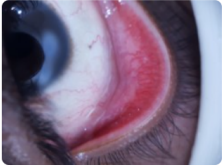
Limit 200MB per file • JPG, PNG, JPEG

Browse files

20200118_164733.jpg

2.4MB

×



Uploaded Eye Image

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2 Clinical Modality

Hemoglobin Level (g/dL)

7.80

−

+

Gender

Male

▼

Age

25

−

+

Country

India

▼

3

Geo-Risk Modality

Select State

Andhra Pradesh

▼

Select District

West Godavari

▼

🔍 Predict Anemia Risk

AnemiaFusionNet

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AnemiaFusionNet — Result

Final Prediction: **High Risk**

Transformer Fusion Score: 0.943

> [View Modality-wise Breakdown](#)

Recommended Precautions & Health Habits

Your indicators suggest a high likelihood of anemia. Please consult a healthcare professional promptly.

- Consult a doctor or hematologist as soon as possible for a confirmatory blood test (CBC).
- Do not self-medicate with iron supplements without medical supervision.
- Prioritize iron-rich, easily absorbed foods: red meat, liver, spinach, lentils, fortified cereals.
- Treat possible underlying causes — heavy menstrual bleeding, GI issues, or chronic illness — with a doctor's guidance.
- Avoid strenuous physical exertion until your hemoglobin levels are evaluated.

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This golden retriever...

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AnemiaFusionNet

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Deploy

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- Treat possible underlying causes — heavy menstrual bleeding, GI issues, or chronic illness — with a doctor's guidance.
- Avoid strenuous physical exertion until your hemoglobin levels are evaluated.
- Watch for warning signs: severe fatigue, rapid heartbeat, breathlessness, pale skin/nails — seek urgent care if these occur.

⚠ This is an AI-based screening estimate, not a medical diagnosis. Please consult a certified healthcare professional for confirmation and treatment.

[Take Assessment Again](#)

[Back to Home](#)

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12. Project Achievements

The project successfully achieved:

- ✓ Multimodal AI-based anemia detection
- ✓ CNN-based image analysis
- ✓ Clinical feature processing
- ✓ NFHS-based Geo-Risk analysis
- ✓ Transformer-based feature fusion
- ✓ Interactive Streamlit application

13. Conclusion

AnemiaFusionNet demonstrates the capability of multimodal Artificial Intelligence for healthcare applications.

By combining visual information, clinical parameters, and geographical risk factors, the system provides a more comprehensive approach compared to traditional single-source methods.

The Transformer-based fusion architecture improves the ability to learn relationships between different data modalities and enables region-aware anemia risk prediction.

14. Future Scope

Future improvements include:

- Increasing image dataset size
- Using Vision Transformers for image processing
- Adding real-time medical device integration
- Improving model generalization with larger datasets
- Deploying the application on cloud platforms